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The Editors
npj Materials Degradation
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Dear Editors,

We are pleased to submit our manuscript, “**Sparse data limit what a corrosion model can predict,**” for consideration as an Article in *npj Materials Degradation*.

Mechanistic corrosion models are routinely fitted with four to six parameters to a handful of measurements and then extrapolated across service lifetimes. Whether those parameters are determined by the data is almost never tested, and our central result is that, for a representative case, several of them are not—with consequences large enough to change a lifetime assessment. We establish this for the duplex oxide grown on Nb-stabilized AISI 347 (X6CrNiNb18-10) in simulated boiling-water-reactor hydrothermal water (240 °C, 7 MPa, 0.4 ppm dissolved O₂), using a differentiable, physics-informed inversion of the point defect model (PDM). We believe it is well suited to *npj Materials Degradation*:

- **The first identifiability characterization of a PDM fit.** Profile likelihood, a 1000-member bootstrap, a prior-leverage test and a residual-leverage decomposition are built into the inversion rather than added afterward. They show that of the five parameters of the accepted model, three exposure times leave one undetermined and pin a second only to within an order of magnitude, and that 95% of the fit statistic is carried by three chromium points—80% by a single measurement. These tools are mature in adjacent fields but, to our knowledge, have not been brought to the PDM; the diagnosis transfers to any sparsely sampled degradation model.
- **Consequences large enough to matter.** The identified kinetics fit the calibration data as well as an empirical power law but extrapolate very differently: at ten years the power law predicts a 3.5-to-6.3-fold thicker barrier layer than the mechanistic 535 nm (bootstrap 95% interval 447–704 nm), a prediction varying 1.19-to-1.56-fold across the scanned operating envelope and activation-energy range. A designed exposure beyond roughly 3000 hours would discriminate the two—an actionable recommendation for lifetime assessment, and the reason the re-analysis is worth doing on data already in the literature.
- **A diagnosed and remedied failure mode.** We document a physics-informed inversion failure specific to sparse experimental data—the data loss satisfied while the governing equation is not—and remedy it with a forward-solve self-consistency check, a cautionary and constructive result for the growing use of physics-informed learning in this field.
- **A first mechanistic description for this alloy, honestly bounded.** Prior characterization of AISI 347 oxide in BWR water stopped at empirical per-element power-law fits with no shared physical parameter set; we connect the Cr-rich barrier and Fe-rich outer layers through a single kinetic description. We are explicit that the model structure is not itself Nb-specific, that the neural inversion is validated against an exact classical reference rather than required by the problem, and that the constant-field closure underlying the derived field

strength does not survive a self-consistency test. We would rather state these bounds than have them inferred.

The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. The author declares no competing interests. Supporting code and data are available as described in the manuscript, and the work involves no human or animal subjects. Thank you for considering our submission.

Sincerely,

Conrard Giresse Tetsassi Feugmo